

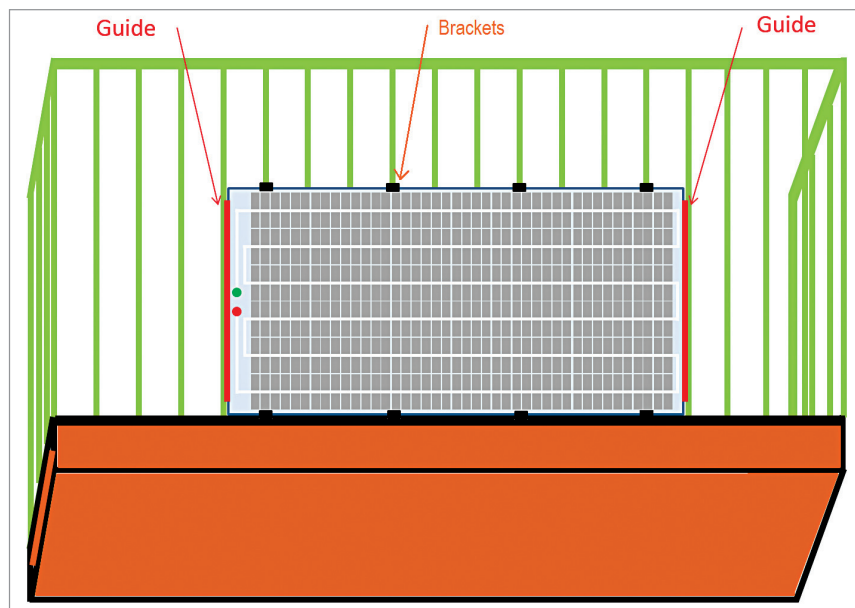


Fig. 3: Polymer Type I panel installed by sliding into the guides

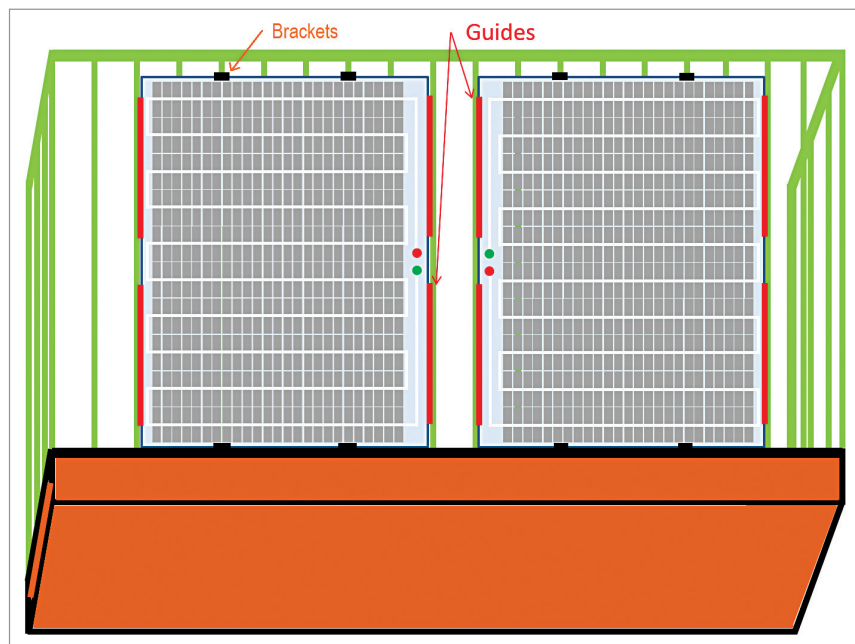


Fig. 4: Polymer Type II panels installed vertically by sliding into the guides

(mono-crystalline) connected in series produces 17.5V at maximum power (V_{mp}). Which means, for generating 120 volts, we have to connect seven such PV panels in series.

PV panel size. As discussed above, heavy solar panels made using glass and aluminium are a safety hazard. Such panels should not be used at all. Luckily, there are solar panels made using polymer. These are light in weight and fairly safe to install. These panels are well suited for apartments.

One drawback of polymer panels is their life. Usually, these last for about fifteen years only, which is acceptable here.

Having standardised the PV voltage to 120 volts, we have to connect seven panels in series. Handling so many panels will become another challenge. One possibility is that all the cells of seven panels be mounted in a single polymer sheet. That means all $36 \times 7 = 252$ cells are mounted inside a single panel and connected in

series. This will simplify the mounting of the panels on the balcony grill. If we select each mono-crystalline solar cell of size $10\text{cm} \times 2.5\text{cm}$ ($4'' \times 1''$), then the panel with 252 such solar cells will produce 100W peak power. Based on this selection criteria, two types of solar panel designs are proposed here.

Panel Type I. Design of Type I panel is shown in Fig. 1. The cells are arranged in a matrix of 42×6 (252 cells). The overall dimensions are $122\text{cm} \times 61\text{cm}$ ($4\text{ft} \times 2\text{ft}$). Due to longer length and smaller width, it is suitable for horizontal mounting.

Panel Type II. Design of Type II panel is shown in Fig. 2. The cells are arranged in a matrix of 25×10 (250 cells). The overall dimensions are $99\text{cm} \times 76\text{cm}$ ($3.25\text{ft} \times 2.5\text{ft}$). This shape and size is suitable for vertical mounting.

Panel mounting

Once the panel sizes are standardised, it becomes easier to install them. When the building is under construction, guides are provided at specified distance on the balcony grills. At the time of construction itself, the builder or customer will decide panel type to be installed.

Fig. 3 shows Type I panel installation on the balcony grills. The builder has already provided two guides spaced 122cm apart, which is the length of the Type I panel. Whenever, user decides to install the panel, he has to just insert it between the two guides. The lower edge of the panel rests on the bottom support provided (during construction). Thus, we see that the mounting is very easy and whenever required the panel can be taken out. (Caution: The user should not do the installation himself. He should employ professionals who have experience in working on high rise buildings.)

Similarly, Fig. 4 shows two Type II panels installed vertically on the balcony grill. In this case the grill has to be slightly taller because the panel height is 99cm. Here, for each panel, four small guides are provided so that the panel fits properly.

Grill vs wall. Some balconies have grills and some have walls. Walls will stop the wind. Hence the panels mounted on the walls are safe against wind. However, through grills wind can blow behind the panel and there is chance that the panel may come out of the guides.